Richard Tsai

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Experience

Business System Analyst 07/2025 – Current

PSEG

- Automated and maintained ETL processes for Electric Transmission & Distribution systems, ensuring integrity, availability, & confidentiality through encryption, access control, and compliance with NIST framework
- Initiated & developed Python-based Relay Generator project to automate SAP data entry, integrating secure coding practices & documenting security protocols in GitHub Wiki for team-wide adherence
- Designed & implemented data migration scripts using SQL and Python, prioritizing data security through role-based access controls (RBAC), encryption at rest/in transit, and audit logging to prevent unauthorized access during migrations
- Automated security patching processes using scripting (Python/Bash), reducing vulnerability exposure by 30%+ & improving system resilience
- Collaborated with security engineers to design secure technical solutions, integrating threat modeling, penetration testing, & secure DevOps practices across system lifecycle
- Streamlined operational tasks with Bash scripting, automating security audits, compliance checks, & log analysis to identify vulnerabilities early and reduce manual effort in daily security operations.

Cloud Application Developer 07/2022 – 08/2024

IBM

- Built full-stack applications using Gatsby, Apollo Client, Java Spring Boot, & Jahia CMS, integrating secure coding practices & DevSecOps principles
- Automated environment setup with Docker and Bash/PowerShell scripts, reducing manual effort by 50% and ensuring secure, reproducible configurations
- Secured CI/CD pipelines on Jenkins using Docker, Terraform Secrets Manager, and Snyk static code analysis, reducing vulnerabilities by 15% through automated code reviews and integration
- Enhanced application performance & security by optimizing Gatsby build processes by 25% & leveraging caching strategies in CI/CD pipelines
- Monitored server logs and Oracle databases with Splunk to detect security incidents, performance bottlenecks, and compliance gaps
- Developed Bash-based security automation scripts to enforce compliance with industry standards and internal policies

Helpdesk Consultant 07/2019 – 05/2022

Rensselaer Polytechnic Institute

- Resolved 90% of 500+ campus network incidents on first contact via Bomgar remote desktop and phone support.
- Documented all customer interactions in Zendesk, ensuring accurate records and efficient follow-ups.
- Implemented a knowledge base system, reducing repeat calls by 50% for recurring technical issues.

Cybersecurity Defensive Operations Programming Intern 05/2021 – 08/2021

The MITRE Corporation

- Contributed as a full-stack developer in IDS/IPS in-house project with PostgreSQL and Redis, Python Flask, and VueJS
- Increased code coverage by 30% through testing using pytest framework, resulting in more reliable, accurate API endpoints
- Improved mean time to detect (MTTD) for security incidents by 20%, allowing for faster response times and reduced damage

Cybersecurity Defensive Operations Intern 06/2020 – 08/2020

The MITRE Corporation

- Analyzed network PCAP files and host-based log data using Zeek and ELK stack dashboard to identify vulnerabilities & remediation efforts
- Automated 90% of Zeek script execution using master bash script, reducing manual effort by 75%
- Implemented custom SIEM dashboard using AngularJS framework, resulting in a 40% reduction in reporting time

Education | Certifications

Rensselaer Polytechnic Institute 05/2022

Bachelor of Science in Computer Science

Certified Information Systems Security Professional (CISSP) 05/2028

CompTIA Security+ 08/2028

AWS Certified Security – Specialty 11/2026

Personal Projects

YACS – Open-Source Schedule Management @ RPI RCOS

Yet Another Course Scheduler – an open-source project to help students plan out their semester

- Led a full-stack project utilizing VueJS, Python FastAPI, and PostgreSQL, containerized with Docker
- Conducted weekly agile sprint meetings, used GitHub Projects to monitor ticket progress, and issued release candidates
- Implemented NGINX best practices to enhance security and maintain optimal performance, conducted CI testing
- Configured Docker containers with secure defaults, including disabling unnecessary services and setting up strict access controls
- Managed GitHub Action CI/CD pipelines and deployment via Kubernetes, ensuring high availability
- Enhanced security on staging servers on OCI, AWS, and on-premises production servers by using Terraform

Personal Home Lab

On-going personal home lab to add more future projects/servers

- Configured FortiGate NGFW with custom IPv4 policies, DoS (Denial-of-Service) policies, VPN, and network segmentation into multiple zones
- Deployed multiple on-home services across servers and containers via Kubernetes clusters with custom networking configurations
- Implemented Grafana, Prometheus and Loki to monitor server activities, network traffic, & detect potential security incidents
- Deployed custom IPS (Intrusion Prevention System) using CrowdSec, which detected & blocked 90% of all incoming threats
- Set up internal recursive DNS servers using Unbound for High-Availability (HA) and DNSSEC validation, serving as a trust anchor for home network

Skills

Languages: Java, Python, C, C++, SQL, Bash, Node.JS frameworks, Typescript

Platforms: Windows, macOS, Redhat Linux, Microsoft Server, AWS, Azure, GCP, Docker, Kubernetes, GitHub, GitLab, BitBucket, Terraform

Tools: Wireshark, NMAP, Metasploit, Hashcat, Zeek, Grafana-Loki-Prometheus stack, ELK stack, Splunk